CONTENT

Prefaţă*……..*I

Chapter 1

PROJECTION SISTEMS……...1

1.1.Central projection system……1
1.2.Parallel projection system ……2
1.3.Axonometrical projection system……4

Chapter 2

PLOTTING OF A POINT……..8

2.1. Double orthogonal projection of a point ……8
2.2. Bisector planes……...10
2.3. Treble orthogonal projection of a point……11
2.4. Points situated in projection planes……...15
2.5. Visibility……15
2.6. Solved problems……15
2.7. Proposed problems……18

Chapter 3

PLOTTING OF A STRAIGHT LINE……20

3.1. Double orthogonal projection of a line……..20
3.2.Traces of the line……21
3.3. Regions of the line……21
3.4. Treble orthogonal projection of a line……...22
3.5. Characteristic positions of a line in report to projection planes……23
3.5.1. Lines parallel to the projection planes……23
3.5.2. Lines perpendicular to the projection planes……...25
3.5.3. Lines laying in a bisector plan ……....26
3.5.4. Lines laying in a projection plane……...27

3.6. The relative positions of a two straight lines……28
3.7. Solved problems……29
3.8. Proposed problems……34

Chapter 4

PLOTTING OF A PLANE……...36

4.1. Plotting of a plane……..36
4.1.1. Line and point laying in a plane……37
4.1.2. Specifying a plane……...38
4.2. The characteristic lines of a plane……40
4.2.1. Lines situated in a plane and parallel to a projection plane……40
4.2.2. The steepest lines of a plane……42
4.3. Characteristic positions of a plane…… 44
4.3.1. Planes parallel to a projection plane. Double projection planes……..44
4.3.2. Planes perpendicular to a projection plane. Projection planes……46
4.4. The relative positions of two planes……..47
4.4.1. Parallel planes…… 48
4.4.2. Intersecting planes……..48
4.5. Relative positions of a line in report to a plane……49
4.5.1. Line lying in a plane……50
4.5.2. Line parallel to the plane……50
4.5.3. Line intersecting a plane……50
4.5.4. Line perpendicular to a plane…… 51
4.6. Visibility in draught……..52
4.6.1. Intersection between a geometrical figure and a line……..52
4.6.2. The intersection of two plane geometrical figures……..52
4.7. Solved problems……53
4.8. Proposed problems……58

Chapter 5

METHODS OF THE DESCRIPTIVE GEOMETRY ……...60

5.1. The method of replacing the projection planes……60
5.1.1. Replacing of a plane for a point……..61
5.1.1.a. Replacing of the vertical plane for a point……61
5.1.1.b. Replacing of a horizontal plane for a point……...61
5.1.2. Replacing of a plane for a line……62
5.1.2.a. Replacing of the vertical plane for a line……...62
5.1.2.b. Replacing of a horizontal plane for a line……..63
5.1.3. Replacing of a plane for a plane……..63

5.1.3.a. Replacing of the vertical plane for a plane……63
5.1.3.b. Replacing of the horizontal plane for a plane……64
5.2. The method of revolution……..65
5.2.1. The level revolution……65
5.2.1.a. The level revolution of a point……...65
5.2.1.b. The level revolution of a line……66
5.2.1.c. The level revolution of a plane……...67
5.2.2. The frontal revolution……..68
5.2.2.a. The frontal revolution of a point ……..68
5.2.2.b. The frontal revolution of a line……..68
5.2.2.c. The frontal revolution of a plane...…….69
5.3. The folding method……...70
5.3.1. Folding on the horizontal projection plane……70
5.3.2. Folding on a level plane……..72
5.3.3. Folding the projection planes ……73
5.3.4. Raising the folding……..74
5.4. Solved problems ……...75
5.5 Proposed problems ……82

Chapter 6

METRIC PROBLEMS. FINDING AND CONSTRUCTION OF DISTANCES AND ANGLES…….84

6.1. Determining the true size of a line-segment and of the angles of inclination to the projection planes……..84
6.2. Determining the distance from a point to a line……87
6.3. Drawing through a point a line-segment inclined at given angles to the projection plane……...88
6.4. Drawing a line intersecting a line-segment on a given angle……90
6.5. Determining the distance between two parallel lines……91
6.6. Determining the distance from a point to a plane……..93
6.7. Determining the true size of the angle between two line-segment……95
6.8. Construction of the bisectrix of an angle of a triangle……96
6.9. Solved problems……97
6.10. Proposed problems……100

Chapter 7

PLOTTING OF THE SOLIDS……...103

7.1. Polyhedra…….103
7.1.1. General knowledge. Apparent outline……...103
7.1.2. Regular polyhedra…… 108
7.1.3. Plane sections through polyhedra.
The development of the polyhedral surfaces……...110
7.1.3.a. Plane sections with projecting planes..……111
7.1.3.b. Plane sections with common planes……115
7.1.4. The intersection of a polyhedron with a line...……..119
7.1.4.a. The intersection of a prism with a line…… 120
7.1.4.b. The intersection of a pyramid with a line……....121
7.2. Cylindrical-conical surfaces……... 122
7.2.1. Plane sections through cylindrical-conical surfaces…… 122
7.2.1.a. Section through cylinder with a common plane……...122
7.2.1.b. Section through a cone with characteristic planes.
The theorem of Dandelin……..124
7.2.1.c. Section through a cone with a common plane…….127
7.3. Solved problems……..130
7.4. Proposed problems……..138

Chapter 8

INTERSECTIONS OF SOLIDS……139

8.1.Polyhedra intersections…….139
8.1.1. Intersection prism– prism……..140
8.1.2. Intersection prism – pyramid…….144
8.1.3. Intersection pyramid – pyramid……146
8.2. Intersections of through cylindrical-conical surfaces……..149
8.2.1. Intersection cylinder – cylinder…….149
8.2.2. Intersection cone – cone……151
8.2.3. Intersection cone – cylinder……..153
8.3. Solved problems ……154
8.4. Proposed problems ……164

Chapter 9

SPHERA. TORE……..165

9.1. Sphera……..165

9.1.1. General knowledge . Plotting. Point on a surface.
The tangent plane in a point on a surface……165

9.1.2. Plane sections through sphere……...168

9.1.2.a. Sections through sphere with a projecting plane…….168

9.1.2.b. Section through sphere with a common plane…….169

9.1.3. Intersection of a line with a sphere……170

9.1.4.The approximate development of the sphere…….173

9.2. Tore……..174

9.2.1 General knowledge . Plotting. Point on a surface.
The tangent plane in a point on a surface……174

9.2.2. Plane sections through tore……...177

9.2.2.a. Sections through tore with a projecting plane …….177

9.2.2.b. Section through tore with a common plane…… 178

9.2.3. Intersection of a line with a tore……180

9.3. Solved problems……..181

9.4. Proposed problems……..184

Chapter 10

QUOTED PROJECTION……...185

10.1. Plotting of a point……..185

10.2. Plotting of a line……186

10.2.1 Characteristic positions of the line……188

10.2.2. The relative positions of two straight lines……188

10.3. Plotting of the plane……189

10.3.1. The positions of a point in report to a plane……. ……190

10.3.2. Characteristic positions of a plane……..……190

10.3.3. The relative positions of two planes ……..……191

10.3.4. Characteristic positions of the line in report to a plane……..……192

10.4.Application ……195

10.5. Solved problems. ……...196

10.6. Proposed problems ……..…..197